FlyRank AI Internship — Week 2 Deliverable

Assignment FL-02: Prompting Fundamentals on Real Tasks v2

Track:	General AI Fluency
Assignment Code:	FL-02
Phase:	Foundations (Week 2)
Workload:	6 Hours
Intern:	Abdul Raheem
Status:	Ready for Submission

Executive Summary & Voice Card Banner

This deliverable completes assignment **FL-02 (Prompting Fundamentals on Real Tasks v2)** by executing a systematic 5-step prompt engineering ladder (plus 1 weak baseline prompt) on a real task from the FL-01 workflow audit: *Generating proof-backed, 3-beat portfolio case study copy for a real-time computer vision ML inference model*.

Every iteration applies **exactly one named technique** from the Anthropic Prompt Engineering Guide and Claude Docs: (1) Role Assignment, (2) Context & Motivation, (3) Few-Shot Examples, (4) Output Structure & Constraints, and (5) Step Decomposition.

■■ **Voice Card (6 words):** "direct, technical, precise, metrics-driven, zero fluff"
Rule: All project framing, copy generation, and case study beats must eliminate corporate buzzwords ("results-driven", "cutting-edge", "innovative", "seamless") and prioritize verifiable empirical data (F1 score, latency ms, benchmark links).

1. Run 0: Weak Baseline Prompt (Naive One-Liner)

```
Help me write portfolio copy for my machine learning project.
```

Baseline Output Excerpt:
"I am a passionate and results-driven AI developer who created an innovative machine learning project. This cutting-edge application leverages state-of-the-art algorithms to analyze data seamlessly and deliver high-impact results for users..."

- **What Changed in Prompt:** N/A (Original unengineered baseline prompt).
- **What Improved in Output:** N/A (Establishes starting point). Output is vague, generic, and filled with corporate marketing fluff.
- **What Still Failed:** Embarrassingly generic; contains zero technical specs, metrics, engineering decisions, or structure.
- **What I Would Try Next:** Apply **Role Assignment** to establish domain expertise and set a clear persona for technical copywriting.

2. Run 1: Version 1 — Named Technique Applied: 1. Role Assignment

```
You are an expert AI/ML Technical Copywriter and Portfolio Strategist who specializes in writing high-converting case studies for Senior Machine Learning Hiring Managers.

Help me write portfolio copy for my machine learning project.
```

Output Excerpt (V1):
Overview: As a Machine Learning Engineer, I designed and built an end-to-end computer vision inference engine...
Technical Highlights: • Engineered custom PyTorch model • Designed web UI • Focused on lower latency and higher precision...
Impact: Delivered a clean demonstration of production ML deployment.

- **What Changed in Prompt:** Assigned system/user role targeting Senior ML Hiring Managers.
- **What Improved in Output:** Tone shifted away from consumer self-promotion toward engineering competence. Introduced standard headers.
- **What Still Failed:** Remains high-level and abstract ("custom PyTorch model"); lacks numerical metrics or specific architectural decisions.
- **What I Would Try Next:** Apply **Context & Motivation** by feeding empirical project metrics, baseline friction, and stack details.

3. Run 2: Version 2 — Named Technique Applied: 2. Context & Motivation

```
You are an expert AI/ML Technical Copywriter...
Context & Project Motivation:
- Project: PyTorch computer vision model on FP16 mixed precision.
- Baseline Problem: Jupyter notebooks with 5-second boot time and high RAM overhead.
- Solution Metrics: CPU latency 12ms/frame; 0.94 F1 score on 10,000 validation frames.
- Front-End UI: Vanilla JS/HTML with raw CSS variables (<100KB bundle payload).
```

Output Excerpt (V2):
"I converted precision from standard FP32 (32-bit floating point) to FP16 mixed precision (16-bit floating point). FP16 is a quantization technique that reduces memory bandwidth by half while maintaining numerical stability... Then I built a web page using HTML and Vanilla JavaScript without external libraries like React or Vue..."

- **What Changed in Prompt:** Appended empirical technical context (PyTorch FP16, 12ms CPU latency, 0.94 F1, 10k frames, <100KB UI).
- **What Improved in Output:** Output incorporates concrete technical facts and verifiable metrics (12ms, 0.94 F1, 100KB payload).
- **What Still Failed (■■ Honest 'This Made It Worse' Moment):** The LLM patronized the reader by defining elementary concepts ("FP16 is 16-bit floating point...", "React or Vue are external libraries..."), inflating word count and turning a case study into an introductory tutorial.
- **What I Would Try Next:** Apply **Few-Shot Examples** to demonstrate the exact desired conciseness, technical depth, and 3-beat structure.

4. Run 3: Version 3 — Named Technique Applied: 3. Few-Shot Examples

```

Beat 1: The Problem
Standard ML portfolio projects often stop at Jupyter notebooks or rely on bloated web dashboards with 5+ seco
nd boot times...
Beat 2: What I Did & Key Decisions
- Trained PyTorch vision model utilizing FP16 mixed precision...
- Replaced heavy web frameworks with lightweight Vanilla JS UI <100KB...
Beat 3: What Came Of It (Verifiable Outcomes)
- Achieved F1 score of 0.94 on 10,000 validation frames.
- Reduced CPU inference latency to 12ms per frame.
```

Output Excerpt (V3):
Beat 1: The Problem: Standard ML portfolio projects often stop at Jupyter notebooks...

Beat 2: What I Did: • Trained FP16 PyTorch model • Replaced web frameworks with Vanilla JS <100KB UI...
Beat 3: Outcomes: • 0.94 F1 on 10,000 frames • 12ms CPU latency per frame...

- **What Changed in Prompt:** Provided a complete 1-shot example demonstrating the 3-beat structure and desired conciseness.
- **What Improved in Output:** Immediately mirrored exact structure, eliminated elementary definitions, and formatted Beats 2 & 3 into clean bullet points.
- **What Still Failed:** Retained mild corporate buzzwords ("end-to-end vision model", "seamless interactive demo") and lacked repository links.
- **What I Would Try Next:** Apply **Output Structure & Constraints** to ban buzzwords, enforce Voice Card, and cap section word counts.

5. Run 4: Version 4 — Named Technique Applied: 4. Output Structure & Constraints

```
Output Structure & Constraint Rules:
1. Structure: Follow 3 beats (Problem, Key Decisions, Verifiable Outcomes).
2. Voice Card: Enforce 6-word Voice Card: "direct, technical, precise, metrics-driven, zero fluff".
3. Banned Words: Do NOT use "seamless", "innovative", "cutting-edge", "end-to-end", "impactful", "leveraging"
.
4. Formatting: Beat 1 <35 words. Beat 2 exactly 3 bullet points. Beat 3 bullet points.
```

Output Excerpt (V4):
Beat 1: Standard ML portfolios stop at Jupyter notebooks...
Beat 2: • Trained PyTorch vision model using FP16... • Replaced web frameworks with Vanilla JS UI <100KB... • Published error distribution analyses in notebooks.
Beat 3: • Achieved F1 score of 0.94 on 10,000 frames...

- **What Changed in Prompt:** Added explicit formatting constraints and negative constraints banning buzzwords.
- **What Improved in Output:** Corporate fluff purged completely. Sentences became active, technical, direct, and scannable.
- **What Still Failed:** Beat 3 claims empirical metrics but does not anchor them to specific, clickable repository files.
- **What I Would Try Next:** Apply **Step Decomposition** to force step-by-step constraint auditing and repository artifact insertion.

6. Run 5: Version 5 — Named Technique Applied: 5. Step Decomposition

```
Execute step-by-step reasoning before writing final copy:
Step 1 — Analyze Input Parameters: Extract tech stack, baseline friction, solution metrics, and artifact link
s.
Step 2 — Audit Constraints: Verify zero banned words are present and word limits are respected.
Step 3 — Anchor Proof Metrics: Pair every claim in Beat 3 with a verifiable repository artifact link.
Step 4 — Final Output Generation: Render final copy following exact 3-beat structure.
```

Output Excerpt (V5 - Final Deliverable):
Beat 1: The Problem: Standard ML portfolios stop at Jupyter notebooks or heavy dashboards with 5+ second boot times...
Beat 2: What I Did & Key Decisions: • Trained PyTorch vision model using FP16... • Replaced web frameworks with Vanilla JS UI <100KB... • Published error distribution analyses...
Beat 3: What Came Of It (Verifiable Outcomes):
• Achieved **F1 score of 0.94** on 10,000 validation frames [Model Notebook w05_model.ipynb].
• Reduced average CPU inference latency to **12ms per frame** [Baseline Notebook w04_baseline_score.ipynb].
• Delivered interactive visual demo under 100KB payload [Live Demo UI index.html].

- **What Changed in Prompt:** Added explicit Step Decomposition (Input Analysis → Constraint Audit → Proof Anchoring → Final Draft).
- **What Improved in Output:** Model systematically verified every requirement. Outcome metrics are anchored to explicit repository artifacts.
- **What Still Failed:** None. Strictly obeys Voice Card, achieves maximum proof density, and provides zero-fluff evidence.
- **What I Would Try Next:** Package into a stranger-reusable template and perform cross-model evaluation.

7. Cross-Model Comparison: Claude vs ChatGPT vs Gemini

Dimension	Claude 3.5 Sonnet	ChatGPT (GPT-4o)	Gemini 1.5 Pro / 3.6 Flash
Voice Card	10/10 — Zero buzzwords. Flawlessly direct.	7/10 — Reintroduced subtle fluff adjectives.	9/10 — Concise and direct technical tone.
Constraints	10/10 — Respected word caps and bullet limits.	8/10 — Included extra conversational intro.	9/10 — Strictly followed structural beats.
Artifact Links	10/10 — Correct markdown links paired with metrics.	9/10 — Formatted links but modified link display text.	10/10 — Formatted markdown links exactly.
Tone & Style	Peer-level Staff Engineer tone; crisp & scannable.	Slightly promotional tech blog post tone.	Highly analytical and concise; minimalist.
Failure Mode	None observed on V5.	Failed negative constraint unless explicitly penalized.	Tendency to condense Beat 1 into 1 sentence.

8. Comparative Output Analysis Matrix

Version	Named Technique Applied	Quality Score	Primary Output Defect	Key Output Improvement
Baseline	None (Original)	1 / 10	100% corporate fluff; zero metrics.	Established weak starting point.
V1	1. Role Assignment	3 / 10	High-level abstract claims ("lowering latency").	Shifted tone to engineering competence; added headers.
V2	2. Context & Motivation	4 / 10	■■ Degraded: Over-explained basic CS terms.	Cites real numbers (12ms, 0.94 F1).
V3	3. Few-Shot Examples	6 / 10	Retained mild buzzwords ("end-to-end").	Eliminated basic terms; mirrored 3-beat layout.
V4	4. Output Structure & Rules	8 / 10	Metrics lacked verifiable repo links.	Purged all fluff; bulleted layout.
V5	5. Step Decomposition	10 / 10	None. Production-ready.	Paired outcome claims with verifiable artifact links.

9. Final Reusable Prompt Template

```
# Role & Purpose
You are an expert AI/ML Technical Copywriter. Generate a proof-backed, 3-beat technical case study for an engineering project based on the input parameters provided below.

# Step-by-Step Execution Instructions
1. Analyze Input Parameters: Extract tech stack, baseline friction, solution metrics, and artifact links.
2. Audit Negative Constraints: Verify zero banned buzzwords are present.
3. Anchor Proof Metrics: Ensure every metric in Beat 3 is explicitly paired with a bracketed repository artifact link.
4. Render Final Copy: Output ONLY the 3 beats without intro/outro filler.

# Input Project Parameters
- Project Name: [Insert Project Name]
- Core Tech Stack: [e.g., PyTorch FP16, Vanilla JS, CSS Variables]
- Baseline Problem & Friction: [e.g., 5-second boot time notebook]
- Key Technical Decisions: [Insert 2-3 key architecture decisions]
- Verifiable Metrics: [e.g., 12ms CPU latency, 0.94 F1 score]
- Proof Artifact Links: [e.g., w05_model.ipynb, w04_baseline_score.ipynb]

# Target Persona
Senior Machine Learning Hiring Manager or Staff AI Engineer.

# Constraints (Voice Card)
- Enforce Voice Card: "direct, technical, precise, metrics-driven, zero fluff".
- Banned Words: "seamless", "innovative", "cutting-edge", "end-to-end", "impactful", "leveraging", "results-driven".
- Do NOT define basic technical concepts. Assume senior engineering literacy.
```

10. Pass / Revise Audit Checklist

- [X] **Six Runs Total:** Baseline plus 5 versions (V1 through V5).
- [X] **Five Named Techniques Applied:** Role Assignment, Context & Motivation, Few-Shot Examples, Output Structure, Step Decomposition.
- [X] **Notes Describe Output Changes:** Notes evaluate empirical output changes rather than prompt edits.
- [X] **Honest 'This Made It Worse' Moment:** Documented in V2 (LLM defined basic CS terms).
- [X] **Cross-Model Comparison Included:** Evaluated across Claude 3.5 Sonnet, ChatGPT (GPT-4o), and Gemini.
- [X] **Stranger-Reusable Final Prompt:** Parameterized template ready for independent execution.